

WBAN System Organization, Network Performance and Access Control: A Review

Israa Al Barazanchi

College of Computing and
Informatics, Universiti Tenaga
Nasional (UNITEN), Malaysia.
israa44444@gmail.com
Computer Engineering

Techniques epartment, Baghdad
College of Economic Sciences
University, Baghdad - Iraq
<https://orcid.org/0000-0002-6798-6295>

Rina Azlin Razali

College of Computing and
Informatics, Universiti Tenaga
Nasional, selangor, Malaysia
Rina@uniten.edu.my

Haider Rasheed Abdulshaheed
Computer Engineering
Techniques Department, Baghdad
College of Economic Sciences
University, Baghdad - Iraq

Wahidah Hashim

College of Computing and
Informatics, Universiti Tenaga
Nasional (UNITEN), Malaysia
Wahidah@uniten.edu.my

Shihab A. Shawkat
University of Samarra , Iraq
orcid.org/0000-0002-9529-2151
shahab84ahmed@gmail.com

Ammar Ahmed Alkahtani

Institute of Sustainable Energy,
Universiti Tenga Nasional,
Kajang, Selangor, Malaysia.
Ammar@uniten.edu.my

Zahraa A. Jaaz

College of Science - Computer
department - AlNahrain
University, Baghdad
zahraa.jaaz@ced.nahrainuniv.edu.iq

Abstract— In the field of Wireless Body Network (WBAN), the network performance of data transmission between the transmitter and the receiving device is one of the most important factors that ensure a healthy system in monitoring the patient's condition. The patient's vital data is transmitted wirelessly through a wave to the receiving devices, which may be a computer or a mobile phone that adopted the use of wireless sensors to monitor the patient's condition. The sensors, usually of high cost, is small with long battery life, must be able to distinguish medical pulses such as electrocardiogram (ECG), photoplethysmogram (PPG), electroencephalogram (EEG), and temperature. Information gathered from the devices in controller will be transferred to remote destinations through remote body-zone system for remedial and analytic goal. This is made possible by the incorporation of wireless network that are meant for long-range transmissions. In this paper presented the basic concept of network performance, sensor node features, introduced an overview for the WBAN application with sensor node design, also explain the Central Control Unit (CCU) and the monitoring of patient system using the CCU. The network topologies with the Sensor communication in the human body explained in this paper.

Keywords—WBAN, Network Performance, wireless network, CCU, data transmission

I. INTRODUCTION

Generally, network performance is the standard that determines the quality of its service in data transmission. Network performance can be defined as a set of tools used to analyze and review network services. In the field of medical environment to determine the quality of the Network performance that transmits patient data to a receiver device, the network performance is analyzed and reviewed over a specified period to measuring and improving the efficiency of this network. Network performance in Wireless Body Network (WBAN) led to the development of efficient transport requirements to enhance network performance like average of

packet loss, end-to-end delays and throughput. These requirements are a measure of the efficiency of network performance and the level of compliance with effective transport requirements. Although there are different studies that have been proposed to improve the efficiency of the network used for the transmission of vital data in WBAN, the current studies does not cover the requirements of effective network performance, as most of them meet energy consumption, MAC protocol and security [1],[2]. WBAN alludes to a gathering of minor energy-efficient remote sensor hubs, which screen human body biomarkers and at the same time choose the dosage of treatment taken by the quiet agreeing to the information gotten [3]. The sensor gadgets ought to be miniaturized, require moo control, and ability to distinguish restorative signals like ECG, PPG, EEG, beat rate, weight, and temperature. The information that collected from the control gadgets was at that point exchanged to inaccessible goals in a WBAN for symptomatic and helpful purposes by joining another remote arrange for long-range transmission [4]. In spite of the fact that the WBAN framework is favorable, the issue of bundle misfortune or debasement through the exchange is still shown. In this way, there's a got to make strides the productivity of full information getting without any misfortune caused by free space impedances of engendering flags like refraction, constriction, assimilation, and reflection. It too depended on the removal between the transmitter and recipient radio wires, the stature and area of the receiving wires, proliferation wave (such as damp or dry discuss), and environment around the receiving wires.

II. WBAN OVERVIEW

A. Wireless body area network

Wireless Body Area Network (WBAN) alludes to a bunch of little brilliantly digital chip node put on the patient body to screen its imperative signals. It gives ceaseless wellbeing observing of an understanding without any limitation on his/her typical lifestyle exercises through the wellbeing care

applications. Due to the type heterogeneous design of the applications, Vital signal rates will vary strongly, extending from basic vital signal at a couple of Kbits/s to a video stream of a few Kbits/s. Vital signal can too be sent in bursts, which suggests that it is sent at the next Vital signal rate amid the bursts. Due to that, the effectiveness of network execution in wellbeing care implementation is one of the foremost critical prerequisites for planning a viable observation implementation Table 1 appears the current benchmarks for on-body and embedded WBANs.

Table 1: Standard for on-body / embedded WBANs [5]

Standard	Frequency band
MICS	402-405 MHz
IEEE 802.15.4	868 MHz Europe and 915 MHz (US)/2.4 GHz
IEEE 802.15.4a	250-750 MHz and 3.1-5 GHz-6-10.6 GHz
UWB	3.1-106 GHz
WMTS	608-1395 MHz and 1400-1427 MHz
IEEE 802.11 a/b/g	2.4 GHz

A WBAN implementation can be described as a group of wired or inaccessible electronic tools that discover together to develop an implementation that checks a patient's basic signals non-stop through sensors on their body. certain types of wave trade patterns that have been assimilated into WBAN systems currently use, in this proposal it shows the basic feature of these kind and shows the feature of distinction among them, although it was one of the modes endorsed from examiner, it has been distinguished based on the efficacy shown in a number of Past considerations. Figure 1 shows the first system of long distance runners.

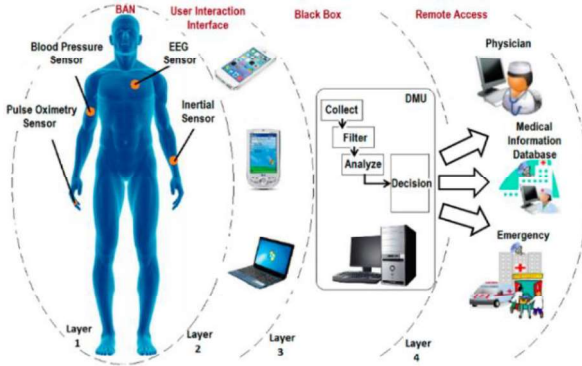


Fig. 1: Configuration of WBAN system [6]

Depending on Figure 1, the most element of WBAN frameworks are sensors, CCU, and recipient gadgets.

B. WBAN applications

The improvement utilize of remote systems going with the nonstop diminishment of electronic gadgets has driven to the advancement of systems of WBAN. WBAN gives farther observing administrations without any limitations and on a persistent premise with the plausibility of doing every day exercises. The foremost utilized applications of the WBAN

innovation within the wellbeing care range are recorded underneath [7],[8].

- **Athlete monitoring:** The WBAN procedures were utilized to degree the developments of competitors precisely in arrange to make strides the execution of competitors to suit the prerequisites of the domain of cutting edge sports. Since the imperative markers depend upon the athletic execution, it is inconceivable for the competitors to be interlinked with wires to the conveyance center or a particular goal in arrange to take after their pointers. Subsequently, the unused WBAN advances permit checking of the imperative markers by the establishment of sensor gadgets completely many points and in ways that suit their development and are broadcast wirelessly to the concerned party.

- **Soldier surveillance :** The WBAN innovation made its way into the war zones through the observing of the injured and wiped out by taking after up their crucial markers. This can be to guarantee appropriate treatment of the troopers as well as to decide those who must be sent to specialized therapeutic centers. In expansion, the WBAN innovation permits early discovery of explosives through temperature and clamor sensors, which might spare the lives of many warriors. Other cases including officers within the front line are the location of warriors who endured from tachycardia or sudden introduction to extraordinary warm at sleep time, which may be the cause of numerous deaths.

- **Advanced animal care:** The sensor innovation makes the observing of imperiled creature species conceivable. This circumstance permits creature masters to keep track of the animals' wellbeing condition whereby the sensors are embedded inside the texture connected to the body of the creatures. The development of the creatures can be observed, whereas other factors such as temperature can be recorded, without causing inconvenience to the creatures, as they will carry on in a more normal development.

III. WBAN SYSTEM COMPONENT

WBAN elements allude to the most component or components of a understanding checking framework in a solid environment. A WBAN framework permits simple internetworking with other gadgets and systems, in this way advertising wellbeing care specialist simple get to the patient's basic and non-critical data. One of the foremost focuses of intrigued of a WBAN is to screen patients remotely utilizing an intranet or the Internet. The WBAN system comprises of two essential parts, program and gear. The program is based on tolerating contraptions, which is chosen by the given company. The hardware comprises essentially of comprises of three essential components that joins sensor centers, a Central Control Unit (CCU) that transmits data to a neighborhood PC and at long final, a getting station at a remedial center. Rough data from a human body is removed by sensor center points. The foremost hardware component in WBAN system comprises of the taking after contraptions:

A. Sensor nodes in WBAN system

Sensor node is planned to gather imperative information that get from patient surface. The detail of the sensor hub in WBAN framework are low in power utilization, size, and reasonably working within the extraordinary sort wave just like the MICS for physiological information preparing. in nearly WBAN framework the sensors node include like equipment element; the sensors collect imperative quiet information. Ordinarily the flag collected from the human body isn't solid and more often than not went with by commotion due to distinctive sound interior in arrange to get ready it to go through an Analog to Advanced Change (ADC) arrange to be changed over into advanced frame for computerized handling. The digitized flag is at that point handled and put away within the chip [9]. The chip will at that point pack those information and transmit them over the discuss by means of a transmitter as delineated in Figure 2.

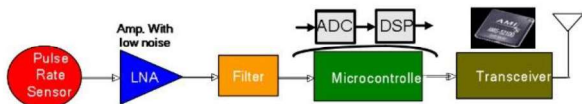


Fig. 2 : Piece chart of a pulse rate sensor hub [10]

The sensors give a persistent wellbeing observing of a persistent without any limitation on his/her ordinary standard of living exercises through the wellbeing care applications. Each sensor hub contains the taking after equipment components (as appeared in Figure 3).

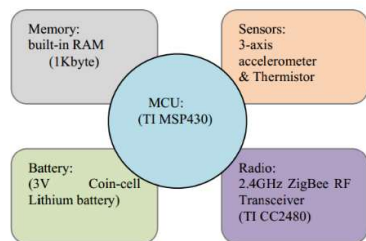


Fig. 3: Hardware components of sensor node [11]

The evaluation of electric sensors and their ensuing preparing for highlight extraction, leads to a collection of real-time physiological factors assembled, which can grant an generally estimation of the user's wellbeing condition at any given time as expressed by [12] as portrayed in Table 2. The table appears diverse sorts of sensor hubs utilized within the WBAN framework with subtle elements of where they are set (in which portion of the body) and work they perform. These sorts are the foremost commonly utilized within the screen of persistent crucial information, there are cases to utilize as it were one or two sorts to screen certain pointers. To plan any observing framework, its imperative to be centered on the quality of the sensors producer and the determinations of the chip and guarantee that all the sensors are consistent with each other to

guarantee recipient gadgets can accepting understanding signals without issues in time of send and get.

Table 2 : Biosensors and biosignals [12]

Type of Bio-signal	Type of Sensor	Description of measured data
Electrocardiogram (ECG)	Skin/Chest electrodes	Electrical activity of the heart (continuous waveform showing the contraction and relaxation phases of the cardiac cycles)
Blood pressure (systolic & diastolic)	Arm cuff-based monitor	Refers to the force exerted by circulating blood on the walls of blood vessels, especially the arteries
Body and/or skin temperature	Temperature probe or skin patch	A measure of the body's ability to generate and get rid of heat
Respiration rate	Piezoelectric/piezoresistive sensor	Number of movements indicative of inspiration and expiration per unit time (breathing rate)
Oxygen saturation	Pulse Oximeter	Indicates the oxygenation or the amount of oxygen that is being "carried" in a patient's blood
Heart rate	Pulse Oximeter/skin electrodes	Frequency of the cardiac cycle
Perspiration (sweating) or skin conductivity	Galvanic Skin Response	Electrical conductance of the skin is associated with the activity of the sweat glands
Heart sounds	Phonocardiograph	A record of heart sounds, produced by a properly placed on the chest microphone (stethoscope)
Blood glucose	Strip-base glucose meters	Measurement of the amount of glucose (main type/source of sugar/energy) in blood
Electromyogram (EMG)	Skin electrodes	Electrical activity of the skeletal muscles (characterizes the neuromuscular system)
Electroencephalogram (EEG)	Scalp-placed electrodes	Measurement of electrical spontaneous brain activity and other brain potentials
Body Movements	Accelerometer	Measurement of acceleration forces in the 3D space

B. CCU in WBAN system

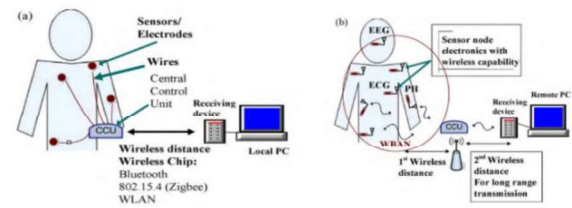
The CCU devise consider main element For any WBAN framework, which accumulates all the transmitted data from sensors within the WBAN and forms the information for observing or demonstrative purposes. Information is exchanged from the RF interface to the CCU through the wave like MICS band once handling has been completed. Following, the information is re-packaged by the CCU some time recently being transmitted to the neighborhood PC. In cases, where restorative centers put at a diverse area, information from the neighborhood PC is transmitted to a farther PC by means of internet or any organize within the therapeutic center. Bluetooth Communication is interfacing the CCU to the versatile gadget, acts as a moment remote connect, which rouses future work. This implies, of communication is still beneath advancement. This remote connect improves patient's portability level, since they can go out to perform every day exercises.

All the information got by the collector station (i.e. the inaccessible PC) can be shown on a Client Show Realistic (GUI). The recipient station can moreover store all the data found within the database framework of a restorative center. CCU is pointed to be worn around a patient's midsection, too depending on quiet portability. Patients following to restricted development, will have the CCU set at get to focus in their houses, inside a remove of 10 meters. Alike the parts of a farther PC at the therapeutic center, the CCU at domestic, is associated to a nearby PC where real-time signal which is getting from the sensor hubs is shown, other than being recorded to a locally set database. Signal which is put away can be conveyed to the database of the therapeutic center, by means of the net. This could be done occasionally. In cases where there's max one in a room (Ex. Clinic), basic program bundles are installed within the CCU and the neighborhood PC. This can be done to modify physiological signals from sensors of each persistent

individually. Data from body are assembled in agreement to time interims that are doled out to patients. Two critical markers that are chosen to screen model framework and to illustrate a more than client arrange are patients' beat rate and temperature [13].

IV. THE ORGANIZATION OF WBAN SYSTEM

The WBAN organize is more often than not organized/located on the human body or exceptionally near to it, whereby more often than not sensors are connected with a fix or found on a wristband. Sensor gadgets are outlined with the conclusion objective of collecting crude signals from the human body, as flag from a human body is customarily weak and joined by commotion. The flag got to involvement escalated and isolating prepare toward start to extend flag qualities, and to arrange of undesirable banners and clamor. The area of hubs on the body and more particularly the way to other node features an expansive affect on the channel quality. As such, a marginally distinctive pose can result in an as of now altogether diverse channel, since the line of locate communication is abruptly incomprehensible. This implies that distinctive individuals will moreover have distinctive channel quality, as the shape of the body moreover impacts the channel [14]. Instep, arbitrary node areas with inexact conceivable WBAN node locations will be produced. This work needs to center on the suggestions of the given channel, instead of attempting to enhance the channel itself. The quality of WBANs channel is the major challenge of organizing, because it will constrain conceivable arrangements. One curiously opportunity is the little scale of the arrange. Patients will most likely don't have hundreds of sensors joined to their body. The application of WBAN in a therapeutic applications may consist of wearable and implantable sensor hubs that recognize natural data from the human body. Other than, it too transmits the data over a brief separate wirelessly to a control gadget, which is either worn on the body or put at a area that's open. The sensor hardware ought to be miniaturized and low-powered. Furthermore, it ought to be able to sense therapeutic signals such as ECG, PPG, EEG, beat rate, weight, and temperature [15]. Information collected from the control gadgets was at that point exchanged to further goals. Usually done by means of WBAN, which is implied for demonstrative and restorative purposes. Other than, it too permits the joining of other remote organizes for long-range transmissions as appeared in Figure 4. Medical centers right now utilize checking gadgets that are not all wearable due to gadgets that are bulky and wires that are utilized for interfacing many sensors. Figure 4(a) depicts the application of a sensor organize that's right now utilized in a certain state of the craftsmanship restorative centers. A CCU collects data from sensors through wires sometimes recently transmitting it to a farther gadget for monitoring purposes. The use of wires in this handle isn't empowered because it brings inconvenience to patients. Other than that, the control unit can be unmanageable as well. Future restorative sensor systems require sensor hubs that are miniaturized and wearable which can communicate wirelessly with the getting gadget as illus.



(a) Current application of medical sensor network (b) Future application of medical sensor network targeted by a WBAN

Fig. 4: A remote sensor organize framework that identifies and transmits signals from a human body [16]

An individual's physiological information just like the heart rate, blood weight, and ECG can be exchanged through a remote connect, which comprises of individual remote sensor hubs. This may be actualized without sending any wired association. Another reality to include is that each sensor has its remote capability and its plan will be optimized with regard to the physical characteristics of the physiological flag [17]. Person remote NODE are a additionally point, as it were certain patients require all the physiological parameters for determination. In future wellbeing care situations, adaptable and stretchable innovation can be utilized to plan sensor hub gadgets which can be implanted in patients' clothing, to make strides the patient's consolation. Two fundamental physical necessities of sensor NODE are moo control operation and miniaturization. These components decide the lifetime of the gadgets as well as their appropriateness for patients. The operation of the remote chip, radio recurrence (RF) transmission and gathering is overwhelmed by the control utilization of sensor hubs. In this manner, a remote stage that gives moo power consumption with least transmitter control is preferable because it is able to meet the specified run of a body-area organize.

V. EXAMPLE OF WBAN SYSTEM

The remote body sensor organize comprises of three primary components that incorporate sensor hubs, a CCU that transmits information to a nearby PC and at last, a accepting gadget at a therapeutic center. Crude information from a human body is extricated by sensor hubs. Information is exchanged from the RF interface to the CCU through the Restorative Embed Communication Benefit (MICS) band once handling has been completed. Following, the signals is re-packaged by the CCU some time recently being send to the nearby receiver device. In cases where restorative centers are found at distinctive areas, information from the nearby PC is transmitted to a farther PC by means of Web or any arrange within the therapeutic center. Bluetooth transmission associated the CCU to the portable gadget, acts as a moment remote interface, which rouses future work. This implies the communication is still beneath advancement. This signal interface improves patient's portability level, since they can go out to perform every day exercises. All the information gotten by the recipient gadget (i.e. the farther PC) can be shown on a Realistic Client Interface (GUI) stage. The collector gadget can too store all the data found

within the database framework of a restorative center. CCU is pointed to be worn around a patient's abdomen, which depends on the patient's portability. Patients following to constrained development will have the CCU put at get to focuses in their houses, inside a separate of 10 m. Alike the parts of a farther PC at the therapeutic center, the CCU at domestic is associated to a neighborhood PC where real-time data gotten from the sensor hubs is shown, other than being recorded to a locally set database. Data that's put away can be conveyed to the database of the restorative center through the Web, which can be done intermittently. In cases where there's more than one persistent living in a room (e.g. the hospital), essential computer program bundles are introduced within the CCU and the nearby PC. Usually finish to modify physiological signals from sensors of each quiet separately. Vital signals is assembled in agreement to time interims that are relegated to body [18].

VI. SENSOR ROLE IN THE PATIENT BODY

In today's globalized society, advancements within the field of hardware, broadcast communications, and particularly coordinates circuit advancement in upgraded observation sensors are essential for the ease of development and to screen imperative signs. These electronic gadgets play a critical part in collecting physiological data from a patient's body and sending it to the restorative center so that the information isn't misplaced, but or maybe arrives at the therapeutic center in a secure and solid way. Hence WBAN has ended up a cutting edge drift within the improvement of electronic circuits and conventions. To include, since the well-being of a quiet, as well as take after ups of their wellbeing issues, have continuously been the point of specialists and therapeutic staff, it is significant to conduct considers recognizing the issues confronted by sensors inside the human body [19]. Remote systems can serve as a communication medium with in-body hubs, e.g. pacemakers. Since of the exceptionally particular channel interior the human body and the exceptionally tall requests with regard to vitality effectiveness, this work limits itself to on-body systems. As outlined in Figure 5, a WBAN comprises numerous hubs found totally different areas of the human body, A Number of gadgets are Connected to the Joints and Appendages, Whereas Sensor D Measures the Heart Rate. The area of a sensor primarily depends on the work of the sensor attached to the hub. In this manner, depending on the desired parameters to screen, diverse sensors and diverse arrange topologies will emerge.

Body Sensor Arrange (BSN) will be utilized to allude to the organize as a uncommon case of Remote Sensor Organize (WSN), where the sensors presently accumulate information on a really little scale as restricted to the classical huge scale WSNs. In addition, there appears to be a drift towards characterizing WBAN inquire about centered on the communication organize whereas BSN investigate will center on preparing the information gotten by implies of a WBAN.

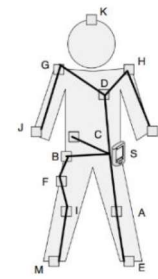


Fig. 5: position of sensors on the human body in WBAN [20]

The IEEE characterizes one other arrange sort close the body as the Wireless Personal Range Arrange (WPAN). The big variance between a WBAN and a WPAN lies within the information rates, whereby WPANs depends technologies like Bluetooth to back maintained rates of up to 3 Mbps. This is often important max the imagined rates from the sensor on the patient body. Subsequently, the WPAN arrange will be shaped within the personal operating space of around 10 m around the body, e.g. between a savvy phone and a printer, whereas a WBAN will be shaped within the body [21].

VII. CHECKING PATIENT STATUS IN MEDICAL ENVIRONMENT

As a rule, when the persistent enters the clinic, the therapeutic group questions him almost his signs. When the reply demonstrates heart malady or those relating to the blood vessels, the persistent will need to remain within the clinic for a certain period of time depending on the tests done upon him. The survival of the persistent within the healing center is taken as a marker of effectively connected sensors to screen crucial markers inside the framework to the specialist and the center of checking patients. Cutting edge frameworks are moving towards the utilize of therapeutic gadgets associated with each other wirelessly whereas guaranteeing effective flag transmission and minimizing the effect of discussing mugginess and impedances of other waves input considering the area of the quiet for the accepting gadgets. There's a got to create arrangements to address the issue of parcel misfortune amid the transmission based on the plan of a framework that guarantees transmission of patient's information within the checking room and exterior without causing alter or misfortune, taking under consideration the patient's consolation and the separate absent from the getting gadgets [22].

VIII. WBAN TOPOLOGIES

For the most part, WBANs have numerous sensor hubs that are scattered around the region of the patient's body and each WBAN particularly has one CCU [23]. The sensor hubs and CCU are orchestrated into one of the two arrange topologies appeared in Figure 6.

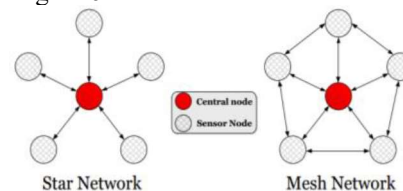


Fig. 6: Network topologies for WBANs [24]

- Star arrange: Inter-node communication does not exist between the sensor hubs and the central hub; pathways are either unidirectional or bidirectional.

- Mesh arrange: A sensor hub has bidirectional communication with another sensor hub and a central hub. Be that as it may, the sensor hubs wirelessly communicate with the CCU but not with another sensor hub, when it is inside a star organize setup. Sensor hubs run on restricted control and computation assets. In this way, a few WBANs utilize unidirectional communication between the sensor hubs and the CCU to protect battery life.

In the meantime, in a work arrange set up, a bigger control source is required at the sensor node's conclusion instep, as sensor hubs communicate to each other than the CCU. As star systems are concerned, small flag handling takes to put on the hubs and crude signals created from the sensors are changed into computerized organize sometime recently continuing to the transmitter piece. In entirety, typically to diminish control utilization, because it brings down computational complexity at the sensor hubs. The transmitted crude flag is at that point passed and analyzed at the CCU, where the control limit is less. Input channel from the CCU to the sensor hubs does not exist, hence synchronization or time-scheduling methods are not conceivable due to the power-constrained nature of the sensor hubs. In this manner, a nonconcurrent WBAN is made whereby sensor hubs transmit information to the CCU at predefined interims which are obscure to other hubs within the arrange [25].

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REFERENCES

- [1] S. S. Tote, S. M. Khupse, and K. S. Bhutwani, "Data Authentication in Wireless Body Area Network (WBAN) Using A Biometric-Based Security," *Int. J. Res. Emerg. Sci. Technol.*, vol. 2, no. 1, pp. 136–142, 2015.
- [2] Z. Zou, Y. Bao, F. Deng, and H. Li, "An Approach of Reliable Data Transmission With Random Redundancy for Wireless Sensors in Structural Health Monitoring," *IEEE Sens. J.*, vol. 15, no. 2, pp. 809–818, 2015.
- [3] D. D. Van, Q. Ai, and Q. Liu, "Vertical Handover Algorithm for WBANs in Ubiquitous Healthcare with Quality of Service Guarantees," *Information*, vol. 8, no. 1, p. 34, 2017, doi: 10.3390/info8010034.
- [4] H. Niu, L. Sun, M. Ito, and K. Sezaki, "Secure Transmission through Multihop Relaying in Wireless Body Area Networks," *IEEE 3rd Glob. Conf. Consum. Electron.*, vol. 2, no. 4, pp. 395–396, 2014.
- [5] A. Ibraheem and M. Manteghi, "Intra-Body Propagation Channel Investigation Using Electrically Coupled Loop Antenna," *Prog. Electromagn. Res.*, vol. 40, no. 57, pp. 57–67, 2014.
- [6] M. Ghamari, B. Janko, R. S. Sherratt, W. Harwin, R. Piechockic, and C. Soltanpur, "A Survey on Wireless Body Area Networks for eHealthcare Systems in Residential Environments," *Sensors (Switzerland)*, vol. 16, no. 6, pp. 1–33, 2016, doi: 10.3390/s16060831.
- [7] I. Al Barazanchi, H. R. Abdulshaheed, M. Safiah, and B. Sidek, "A Survey : Issues and challenges of communication technologies in WBAN," *Sustain. Eng. Innov.*, vol. 1, no. 2, pp. 84–97, 2020.
- [8] H. R. Abdulshaheed, I. Al Barazanchi, M. Safiah, and B. Sidek, "Survey : Benefits of integrating both wireless sensors networks and cloud computing infrastructure," *Sustain. Eng. Innov.*, vol. 1, no. 2, pp. 67–83, 2020.
- [9] I. Al Barazanchi, H. R. Abdulshaheed, S. A. Shawkat, and S. R. Binti, "Identification key scheme to enhance network performance in wireless body area network," *Period. Eng. Nat. Sci.*, vol. 7, no. 2, pp. 895–906, 2019.
- [10] M. R. Yuce, "Implementation of wireless body area networks for healthcare systems," *Sensors Actuators A Phys.*, vol. 162, no. 1, pp. 116–129, Jul. 2010, doi: 10.1016/j.sna.2010.06.004.
- [11] F. Kiyomarsi, "Present A Fuzzy-Logic Based Method for Head-Cluster Selection In Wireless Sensor Networks," *Chem. Process Eng.*, vol. 2, no. 2014, pp. 34–49, 2015.
- [12] A. Pantelopoulous, "Prognosis : a Wearable System for Health Monitoring of People At Risk," *Electrical and Computer Engineering*, University of Patras. [Online]. Available: https://etd.ohiolink.edu/rws_etd/document/get/wright1284754643/inline. [accessed: 16-May-2016]., 2010.
- [13] S. Rajasekaran, P. Kumaran, G. Premnath, and M. Karthik, "Human Health Monitoring Using Wireless Sensors Network (WSN)," *Int. J. Appl. or Innov. Eng. Manag.*, vol. 2, no. 12, pp. 323–330, 2013.
- [14] B. Braem and C. Blondia, "Supporting mobility in Wireless Body Area Networks: An analysis," *Commun. Veh. Technol. Benelux (SCVT), 2011 18th IEEE Symp.*, pp. 1–6, 2011.
- [15] V. Goverdovsky, D. Looney, P. Kidmose, C. Papavassiliou, and D. P. Mandic, "Co-Located Multimodal Sensing: A Next Generation Solution for Wearable Health," *IEEE Sens. J.*, vol. 15, no. 1, pp. 138–145, 2015, doi: 10.1109/JSEN.2014.2338612.
- [16] E. Andreeva, "Alternative Biometric as Method of Information Security of Healthcare Systems," *proceeding 12th Conf. Fruct Assoc.*, vol. 12, no. 17, pp. 20–22, 2012.
- [17] M. Stoppa and A. Chiolerio, "Wearable Electronics and Smart Textiles: A Critical Review," *Sensors (Switzerland)*, vol. 14, no. 7, pp. 11957–11992, 2014, doi: 10.3390/s140711957.
- [18] H. W. Pflug et al., "Radio Channel Characterization for 400 MHz Implanted Devices," *Wirel. Commun. Netw. Conf. (WCNC), 2014. IEEE.*, vol. 1, no. 3, pp. 293–298, 2014.
- [19] S. Ullah, P. Khan, N. Ullah, S. Saleem, H. Higgins, and K. S. Kwak, "A Review of Wireless Body Area Networks for Medical Applications," *arXiv Prepr. arXiv1001.0831*, p. 7, 2010, doi: 10.4236/ijcns.2009.28093.
- [20] B. Braem, "Reliable and Energy Efficient Protocols for Wireless Body Area Networks," *Universiteit Antwerpen (Belgium)*. [Online]. Available: <https://dl.acm.org/citation.cfm?id=2395363.pdf>. [accessed: 12-Feb.-2016]., 2011.
- [21] T. Dou and L. Yi, "The Development and Trend of Wireless Body Area Networks in The Smart Clothing Field," *4th Int. Conf. Mechatronics, Mater. Chem. Comput. Eng.*, no. Icmccce, pp. 1182–1187, 2015.
- [22] Liao et al., "Analysis of In-to-Out Wireless Body Area Network Systems : Towards QoS-Aware Health Internet of Things Applications," *Electronics*, vol. 5, no. 3, p. 38, 2016, doi: 10.3390/electronics5030038.
- [23] Kim et al., "Analysis of Aggregation Delay for Multisource Sensor Data with On-Off Traffic Pattern in Wireless Body Area Networks," *Sensors*, vol. 16, no. 10, p. 1622, 2016, doi: 10.3390/s16101622.
- [24] S. Pathak, M. Kumar, A. Mohan, and B. Kumar, "Energy Optimization of ZigBee Based WBAN for Patient Monitoring," *Procedia Comput. Sci.*, vol. 70, no. 12, pp. 414–420, 2015, doi: 10.1016/j.procs.2015.10.055.
- [25] H. Xiong and Z. Qin, "Revocable and Scalable Certificateless Remote Authentication Protocol With Anonymity," *IEEE Trans. Inf. Forensics Secur.*, vol. 10, no. 7, pp. 1442–1455, 2015.
- [26] H. R. Abdulshaheed, S. A. Binti, and I. I. Sadiq, "A Review on Smart Solutions Based-On Cloud Computing and Wireless Sensing," *Int. J. Pure Appl. Math.*, vol. 119, no. 18, pp. 461–486, 2018.
- [27] I. Al Barazanchi, H. R. Abdulshaheed, M. Safiah, and B. Sidek, "Innovative technologies of wireless sensor network : The applications of WBAN system and environment," *Sustain. Eng. Innov.*, vol. 1, no. 2, pp. 98–105, 2020.
- [28] I. Al Barazanchi, "An Analysis of the Requirements for Efficient Protocols in WBAN," *J. Telecommun. Electron. Comput. Eng.*, vol. 6, no. July, p. 43, 2014.